

USE-CASE FLOW SPECIFICATION

Feature Flag & Dynamic Config Manager | Problem Statement #43

UC-03: Evaluate Feature Flag

Primary actor: Client SDK (external system) **Supporting stakeholder:** Software Engineer

Goal

Determine whether a feature is enabled for a particular user using the flag configuration, targeting rules, and applicable environment settings.

Preconditions

- The feature flag exists and has a unique key.
- The client SDK can send an evaluation request to the platform.
- The user identifier and target environment are available.

Postconditions

- The system returns a boolean feature flag state to the requesting client.
- The result is determined using the applicable targeting rules and environment configuration.

Main Success Scenario

1. The Client SDK sends an evaluation request containing the flag key, user ID, and environment.
2. The system validates the request and identifies the requested feature flag.
3. The system retrieves the flag configuration and the applicable environment override.
4. The system evaluates the user's targeting rules, such as User ID hash assignment or Beta cohort membership.
5. The system determines the final boolean flag state.
6. The system returns the flag state to the Client SDK.

Alternate Flow: Flag Not Found

7. At Step 2, the system cannot find the requested flag key.
8. The system returns a documented not-found response or configured fallback state.
9. The Client SDK handles the response according to its fallback behavior; evaluation ends.

Related requirements: FR-001, FR-002, NFR-001